

Project Description Document

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A. Description of the Data

Data Sources

1. Inflation Data: The dataset was sourced from a CSV file titled `inflationRate.csv`. It contains monthly records of the Consumer Price Index (CPI) as an indicator of inflation.
2. Unemployment Data: The dataset was sourced from a CSV file titled `unemploymentRate.csv`. It provides monthly unemployment rate statistics.

Variables

1. Inflation Data Variables:
 - `Date`: The month and year of the data point (e.g., "2023-06").
 - `ConsumerPriceIndex`: The numerical CPI value.
2. Unemployment Data Variables:
 - `Date`: The month and year of the data point (e.g., "2023-06").
 - `UnemploymentRate`: The percentage rate of unemployment.

Data Cleaning and Processing

To prepare the data for visualization:

- Both datasets contained metadata rows at the top, which were removed using the `skiprows` parameter in Python's `pandas` library.
- Column headers were renamed for clarity, e.g., `Column1` was renamed to `Date` and `Column2` to `ConsumerPriceIndex` or `UnemploymentRate`.
- Invalid dates and null values were removed using `pd.to\_datetime()` for date parsing and `dropna()` to filter out incomplete rows.
- The cleaned datasets were saved as `cleaned\_inflationRate.csv` and `cleaned\_unemploymentRate.csv`.
- The datasets were integrated by aligning their `Date` variables, enabling direct comparison between inflation and unemployment data over time.

B. Visual Design Rationale

Overview

The visualization depicts the relationship between unemployment rate and CPI over time using an interactive line graph. The key design considerations included clarity, usability, and aesthetics to ensure the viewer can derive meaningful insights.

Marks and Channels

1. Lines:

- Unemployment rate: Represented by a steel-blue line.
- Consumer Price Index: Represented by an orange line.
- The lines convey trends over time, using position along the x-axis for dates and the y-axis for values.

2. Points:

- Each data point is marked with a circle for emphasis and interactivity.
- Points are color-coded to match the respective lines.

Axes and Titles

- X-Axis: Represents time, formatted in years for readability.
- Y-Axis: Represents numerical values (percentage for unemployment, index value for CPI).
- Both axes include descriptive titles to reduce ambiguity.

Annotations

- Title: "Unemployment Rate vs Consumer Price Index" is centrally aligned to inform the viewer about the purpose of the graph.
- Tooltips: Dynamic tooltips appear on hovering over points to display specific values.

Trade-Offs

- While both datasets cover extensive time ranges, the visualization prioritizes alignment of overlapping periods for comparison, potentially discarding non-overlapping data.
- The choice of colors and line thickness ensures contrast while avoiding overwhelming visual elements.

C. Interactive Elements and Design Rationale

Interactive Elements

1. Hover Effects:

- Highlight points with additional visual cues (e.g., increased opacity).
- Show emoji icons representing significant points (e.g., highest, second-highest, and lowest values).

2. Toggle Visibility:

- Checkbox controls enable viewers to selectively display or hide individual datasets (e.g., only show unemployment data).

3. Tooltips:

- Provide precise data values and context for hovered points, enhancing comprehension.

Design Decisions

- Discoverability: Interactive elements such as checkboxes are positioned adjacent to the graph for easy access.
- Usability: Hover effects and emojis provide immediate feedback without requiring additional navigation.
- Engagement: Visual cues like fading non-selected lines focus attention on user interaction.

D. The Story

Insights

The visualization highlights:

1. Trends:

- An inverse relationship often exists between unemployment and inflation.
- Economic events (e.g., recessions) manifest as significant peaks or troughs.

2. Surprising Observations:

- The inflation index may rise even during periods of high unemployment, suggesting stagflation.

3. Key Insights:

- By dynamically interacting with the data, viewers can isolate periods of interest, such as economic recoveries or downturns.